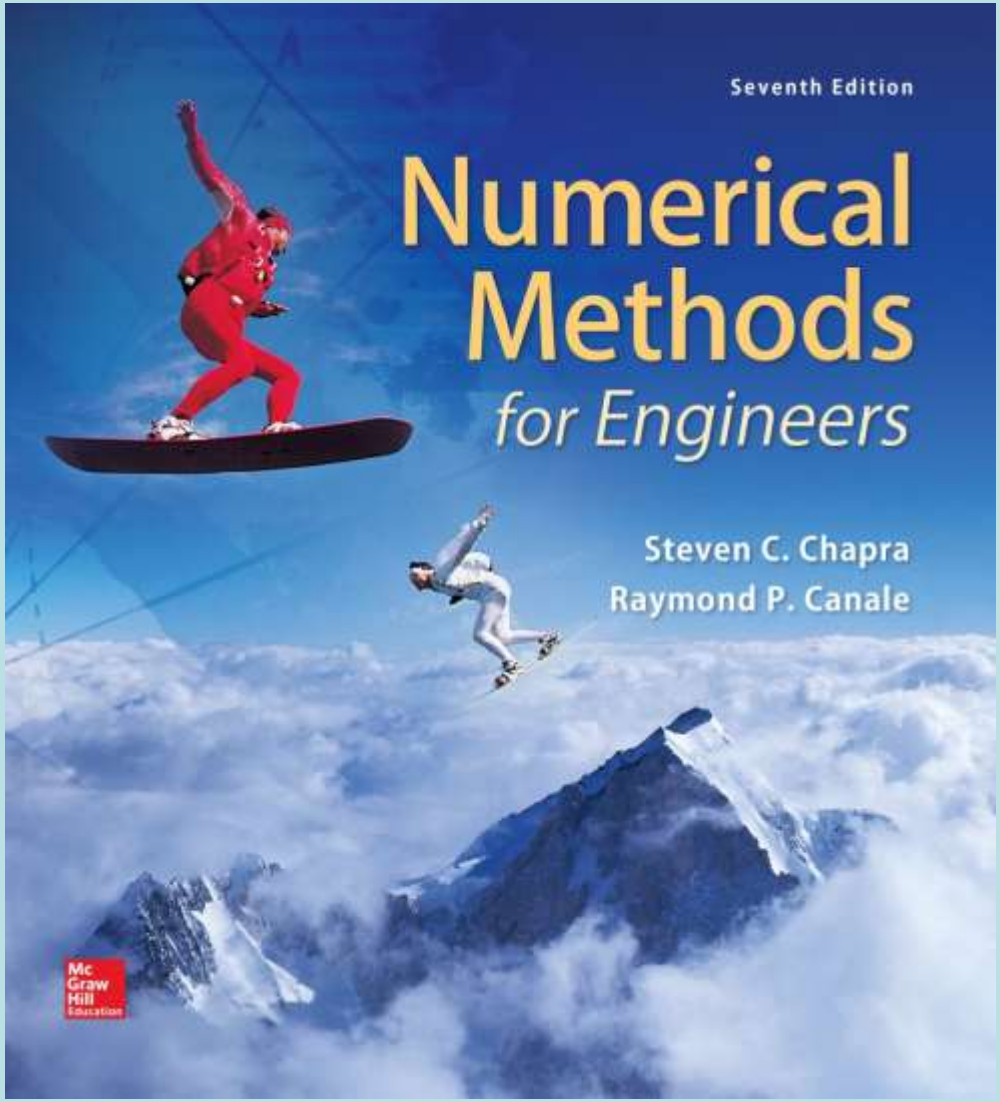
Chapter 14



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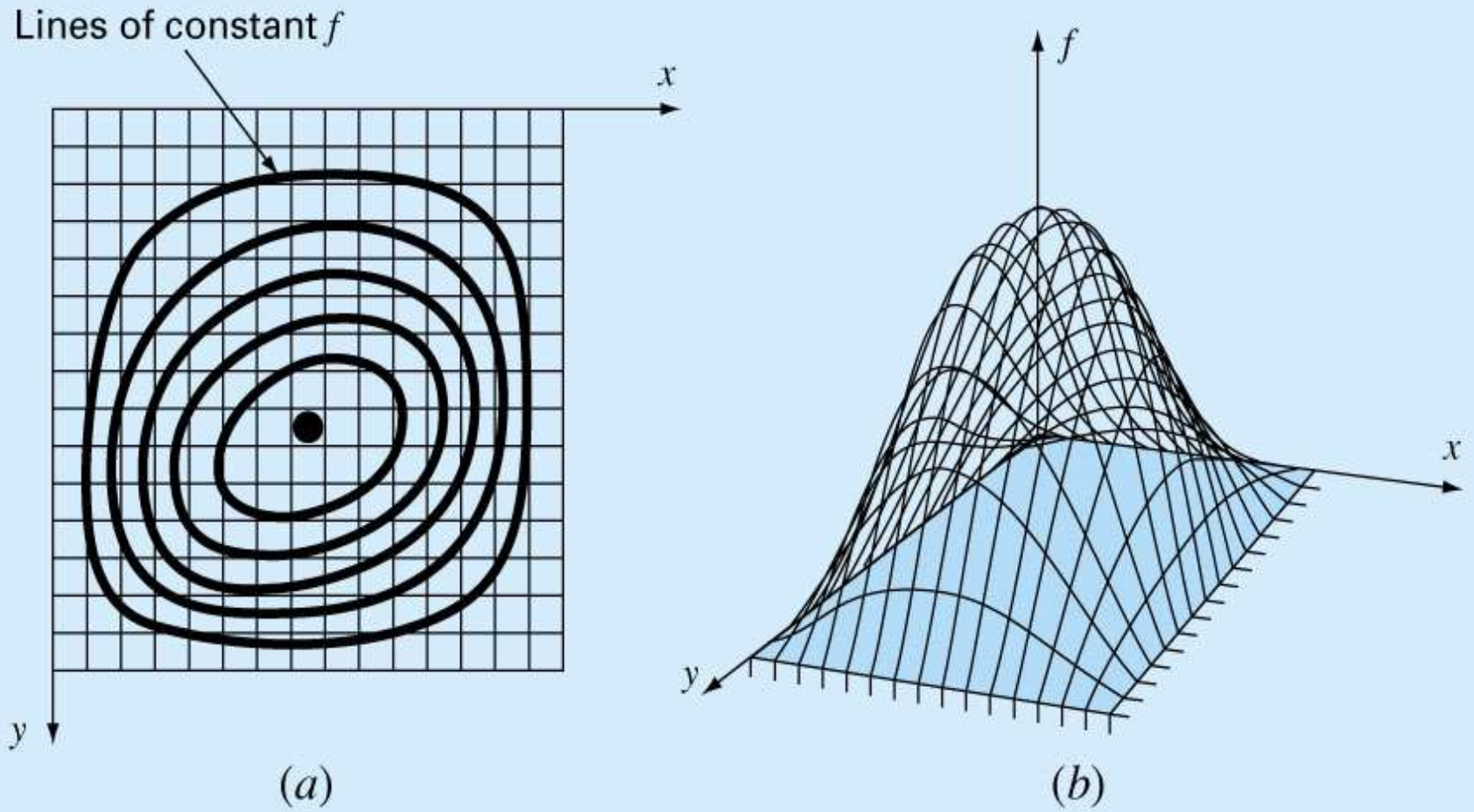
Multidimensional Unconstrained Optimization

Chapter 14

- Techniques to find minimum and maximum of a function of several variables are described.
- These techniques are classified as:
  - That require derivative evaluation
    - *Gradient* or descent (or *ascent*) methods
  - That do not require derivative evaluation
    - *Non-gradient* or *direct* methods.

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Figure 14.1



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DIRECT METHODS
Random Search

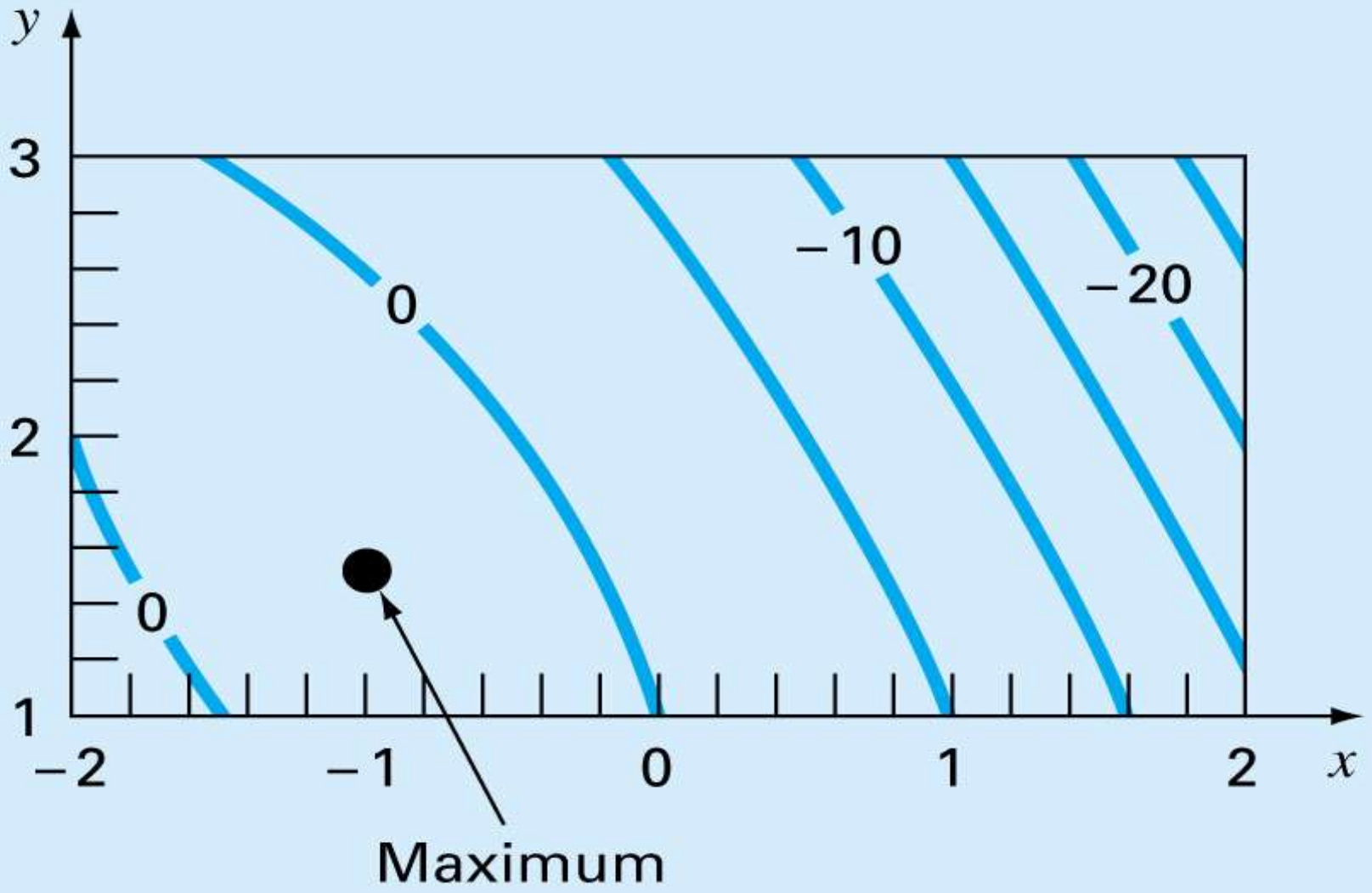
- Based on evaluation of the function randomly at selected values of the independent variables.
- If a sufficient number of samples are conducted, the optimum will be eventually located.
- Example: maximum of a function

$$f(x, y)=y-x-2x^2-2xy-y^2$$

can be found using a random number generator.

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Figure 14.2



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Advantages/

- Works even for discontinuous and nondifferentiable functions.
- Always finds the global optimum rather than the global minimum.

Disadvantages/

- As the number of independent variables grows, the task can become onerous.
- Not efficient, it does not account for the behavior of underlying function.

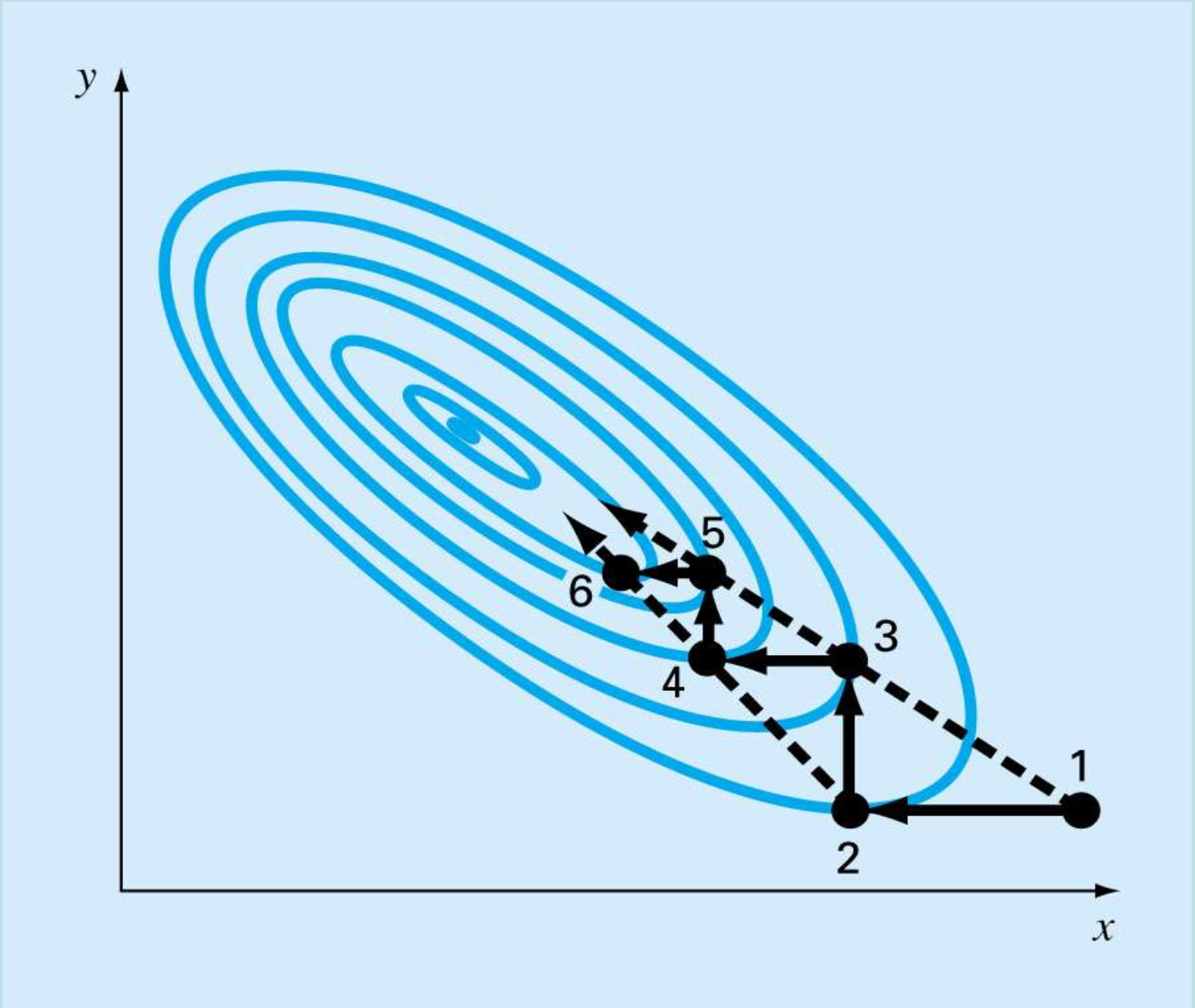
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Univariate and Pattern Searches

- More efficient than random search and still doesn't require derivative evaluation.
- The basic strategy is:
  - Change one variable at a time while the other variables are held constant.
  - Thus problem is reduced to a sequence of one-dimensional searches that can be solved by variety of methods.
  - The search becomes less efficient as you approach the maximum.

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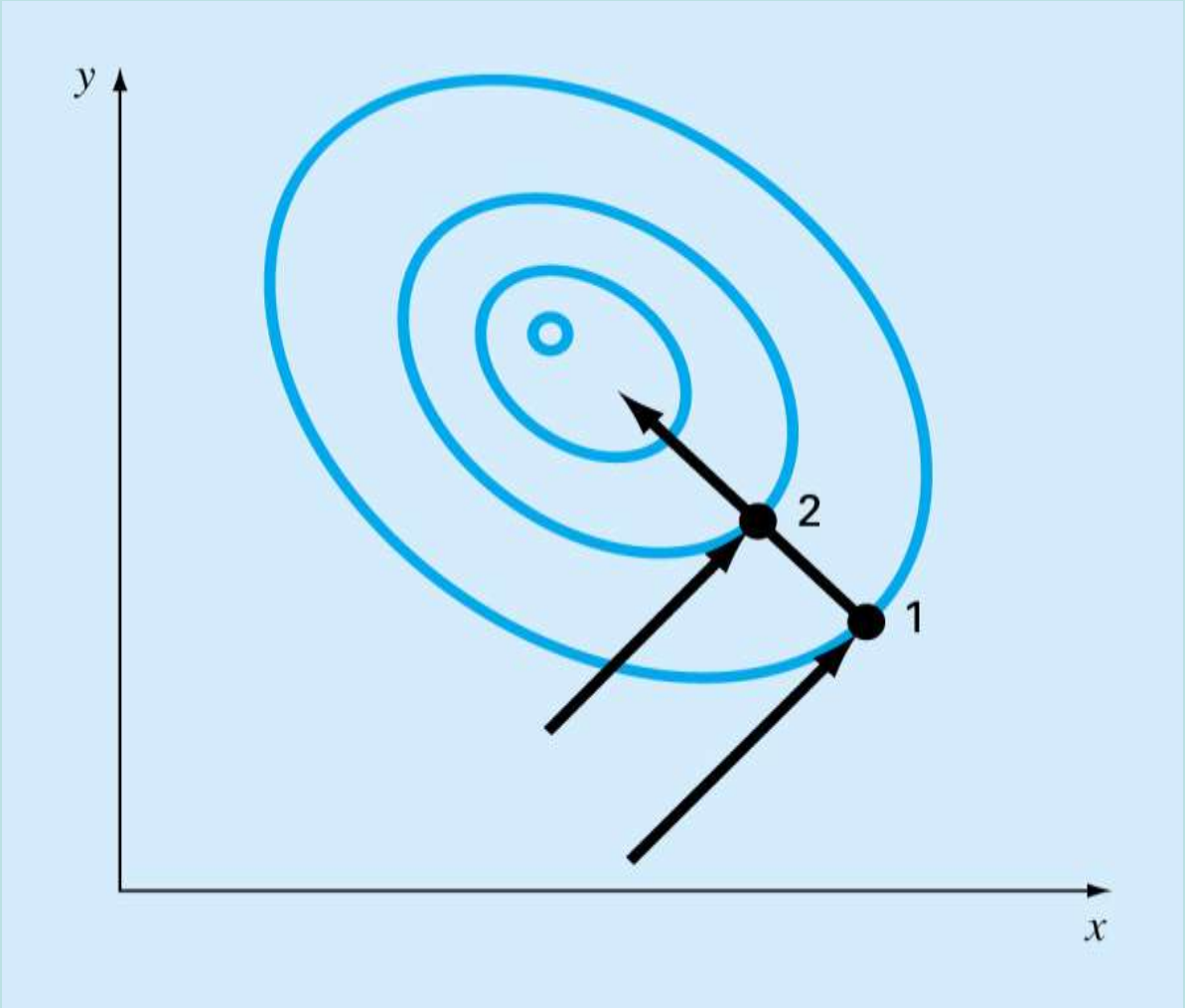
Figure 14.3



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- *Pattern directions* can be used to shoot directly along the ridge towards maximum.

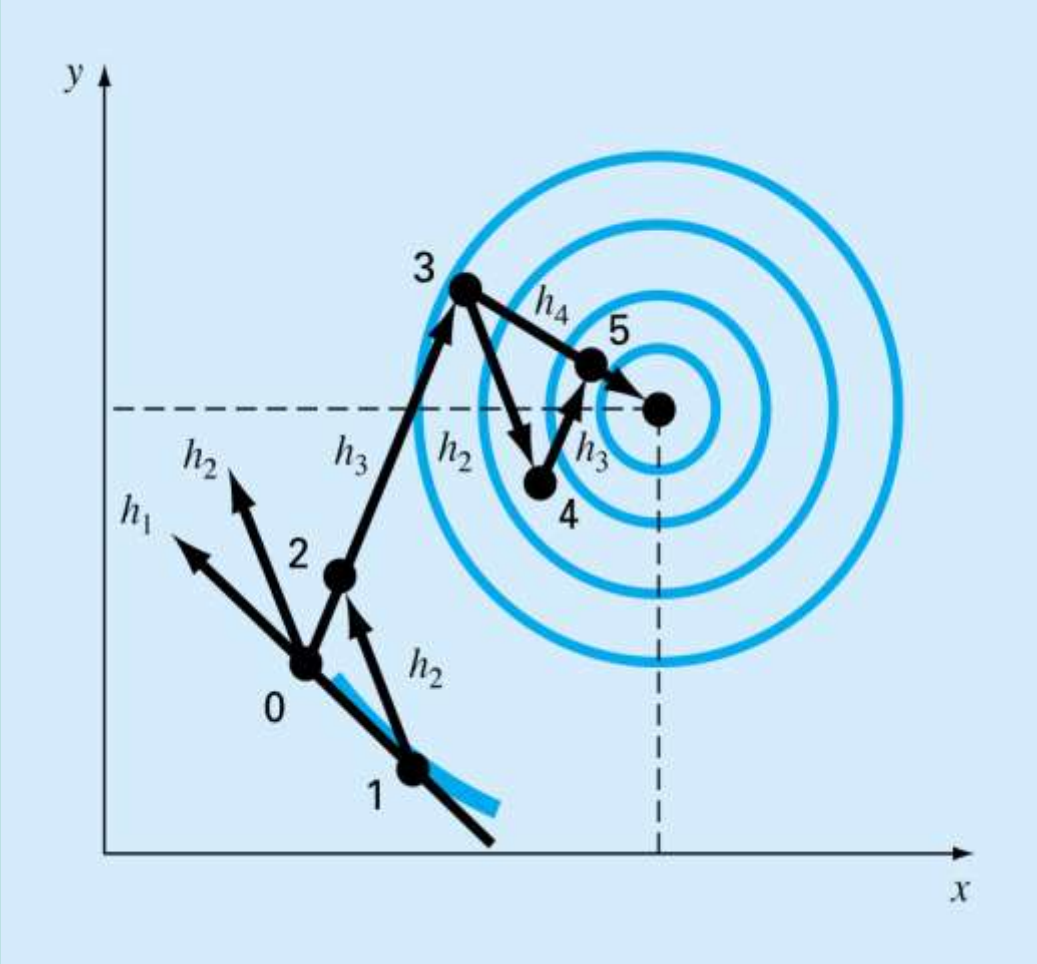
Figure 14.4



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Figure 14.5

- Best known algorithm, *Powell's method*, is based on the observation that if points 1 and 2 are obtained by one-dimensional searches in the same direction but from different starting points, then, the line formed by 1 and 2 will be directed toward the maximum. Such lines are called *conjugate directions*.



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GRADIENT METHODS
Gradients and Hessians

The Gradient/

- If *f(x,y)* is a two dimensional function, the *gradient* vector tells us
  - What direction is the steepest ascend?
  - How much we will gain by taking that step?

$$\nabla f = \frac{\partial f}{\partial x}\,\mathrm{i} + \frac{\partial f}{\partial y}\,\mathrm{j} \qquad \text{or } \textit{del } f$$

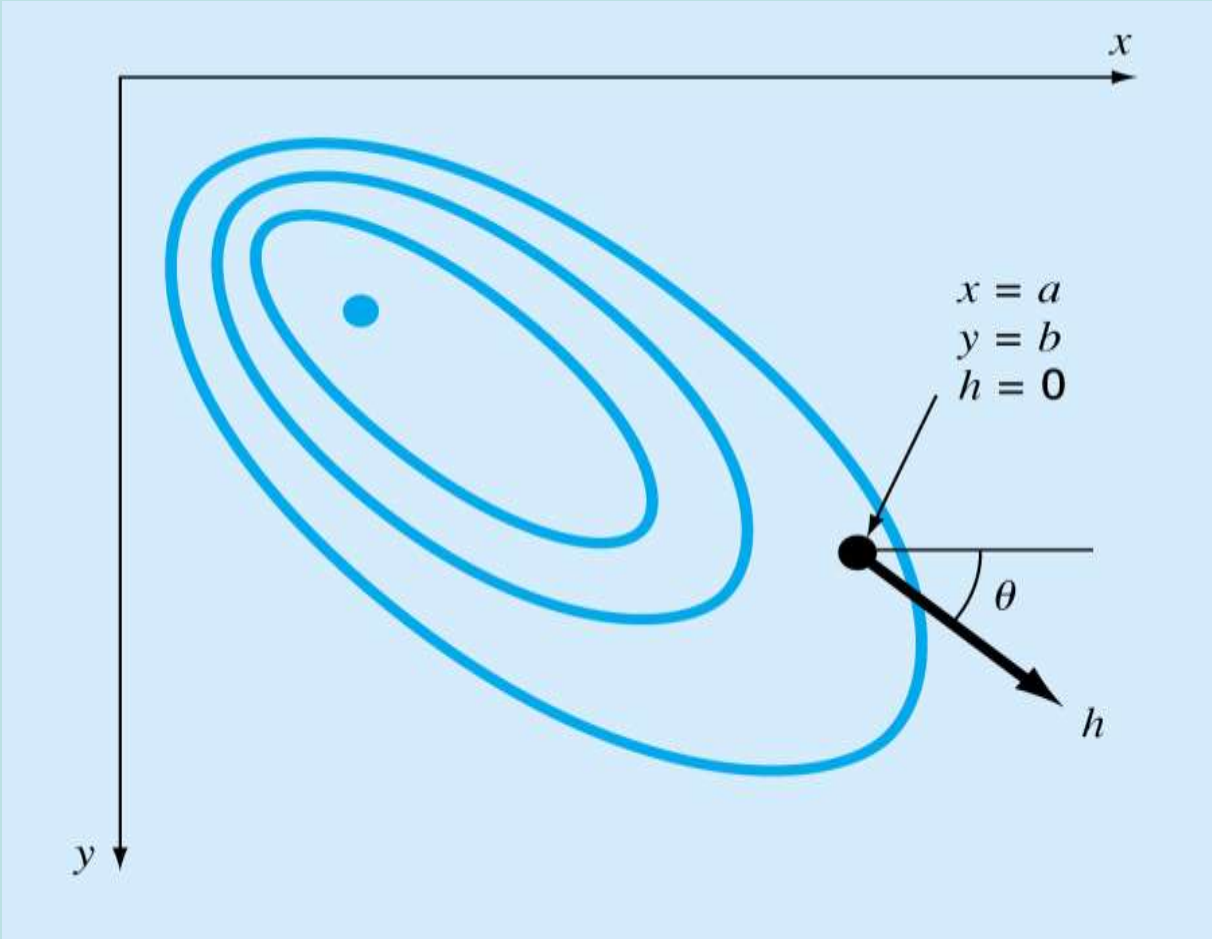
Directional derivative of *f(x,y)* at point x=a and **y=b**

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- For *n* dimensions

$$\nabla f(x) = \begin{Bmatrix} \dfrac{\partial f}{\partial x_1}(\mathrm{x}) \\ \dfrac{\partial f}{\partial x_2}(\mathrm{x}) \\ \vdots \\ \dfrac{\partial f}{\partial x_n}(\mathrm{x}) \end{Bmatrix}$$

Figure 14.6



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The Hessian/

- For one dimensional functions both first and second derivatives valuable information for searching out optima.
  - First derivative provides (a) the steepest trajectory of the function and (b) tells us that we have reached the maximum.
  - Second derivative tells us that whether we are a maximum or minimum.
- For two dimensional functions whether a maximum or a minimum occurs involves not only the partial derivatives w.r.t. x and y but also the second partials w.r.t. x and y.

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Figure 14.7

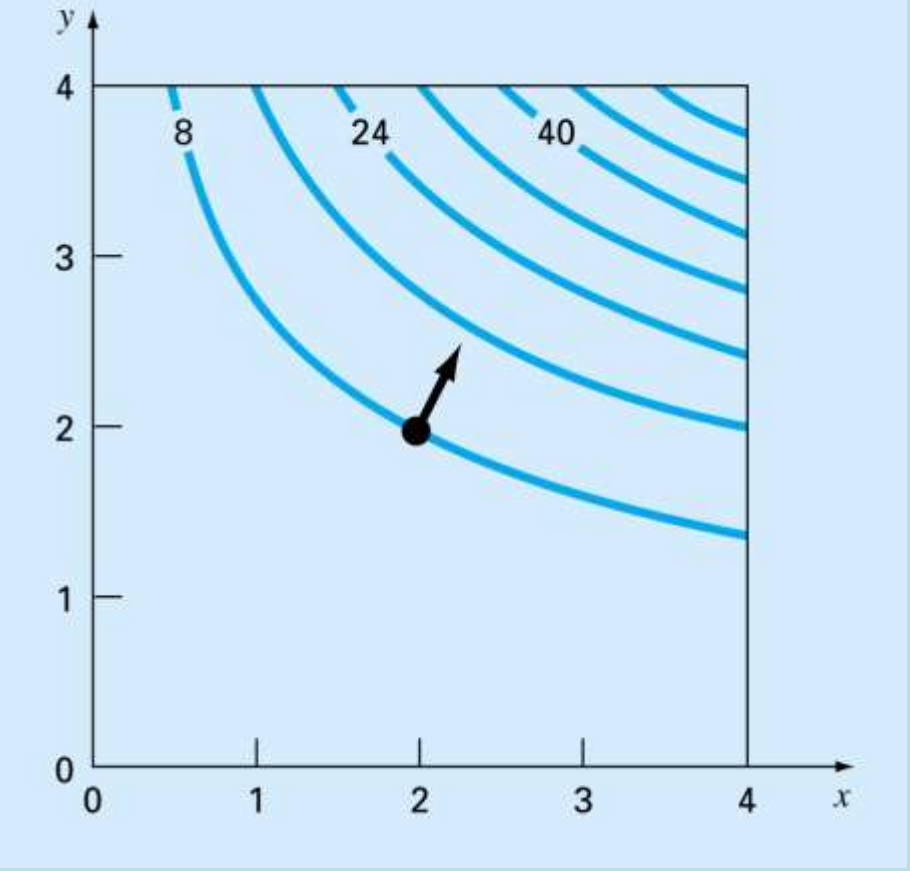
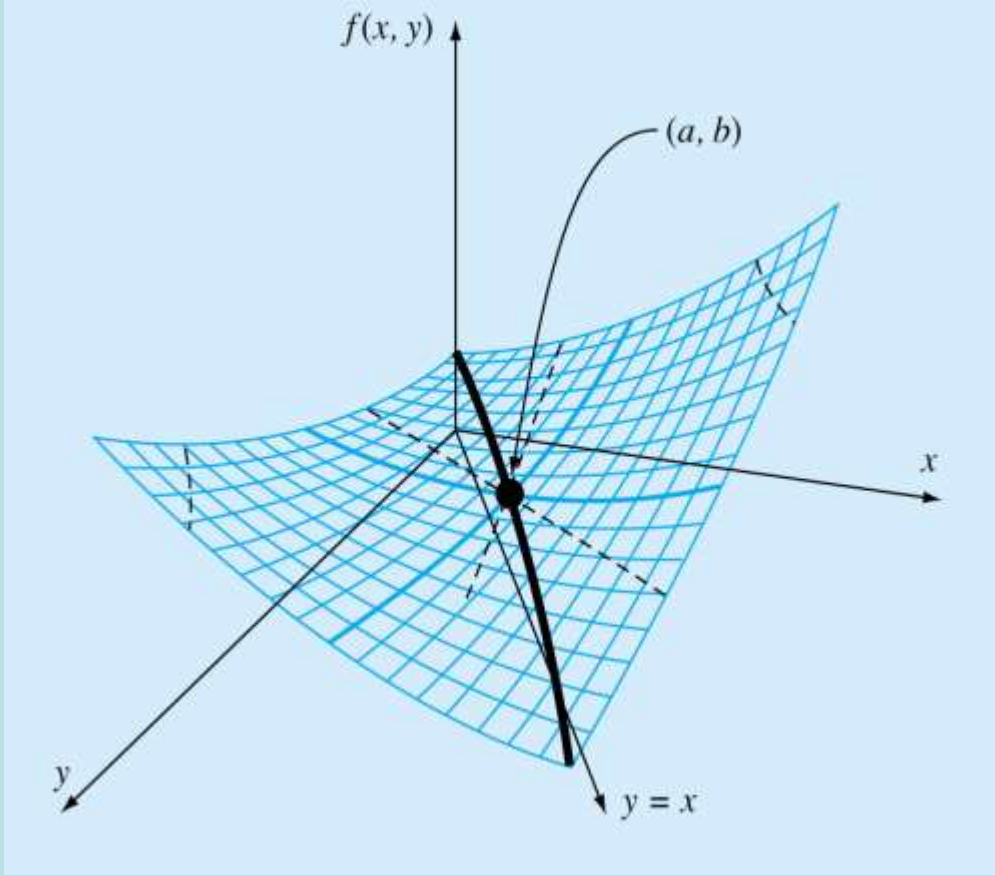


Figure 14.8



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- Assuming that the partial derivatives are continuous at and near the point being evaluated

$$|H| = \frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

Three cases can occur

- If $|H| > 0$ and $\partial^2 f/\partial x^2 > 0$, then $f(x, y)$ has a local minimum.
- If $|H| > 0$ and $\partial^2 f/\partial x^2 < 0$, then $f(x, y)$ has a local maximum.
- If $|H| < 0$, then $f(x, y)$ has a saddle point.

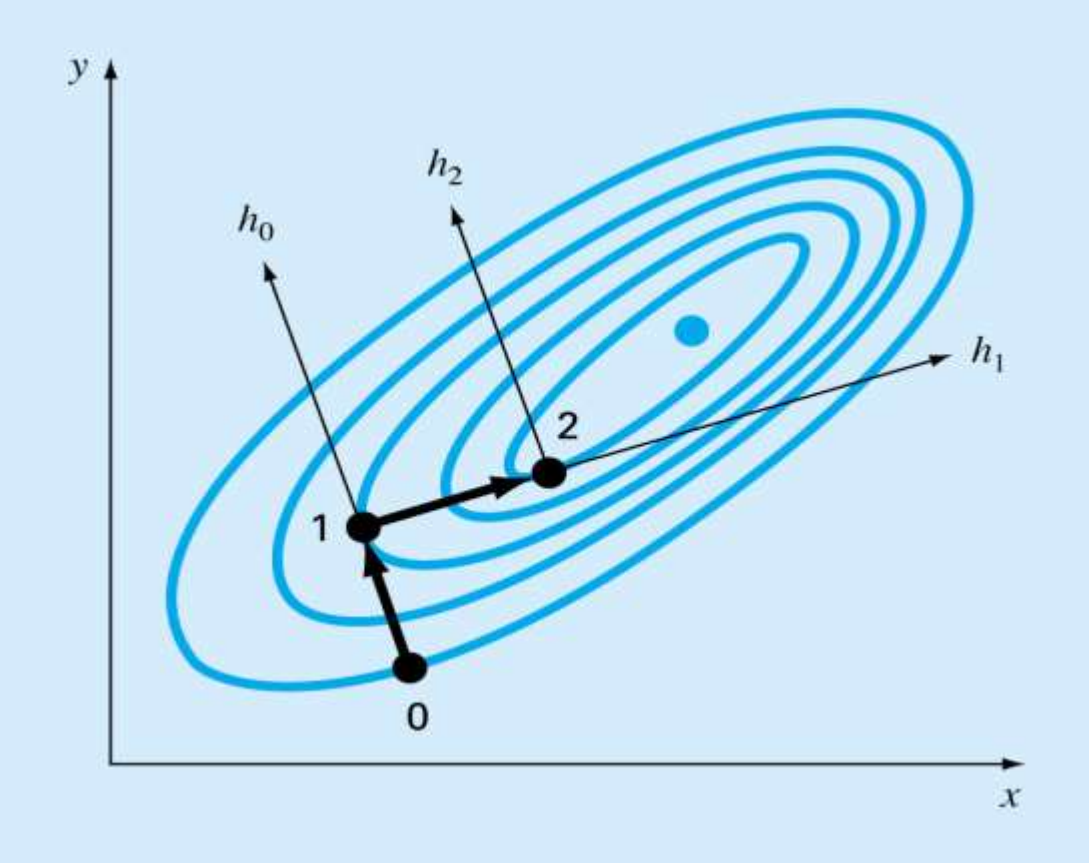
The quantity [H] is equal to the determinant of a matrix made up of second derivatives

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The Steepest Ascend Method

- Start at an initial point (x_o,y_o), determine the direction of steepest ascend, that is, the gradient. Then search along the direction of the gradient, h_o, until we find maximum. Process is then repeated.

Figure 14.9



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- The problem has two parts
  - Determining the "best direction" and
  - Determining the "best value" along that search direction.
- Steepest ascent method uses the gradient approach as its choice for the "best" direction.
- To transform a function of *x* and *y* into a function of *h* along the gradient section:

$$x = x_o + \frac{\partial f}{\partial x} h$$

$$y = y_o + \frac{\partial f}{\partial y} h$$

h is distance along the h axis

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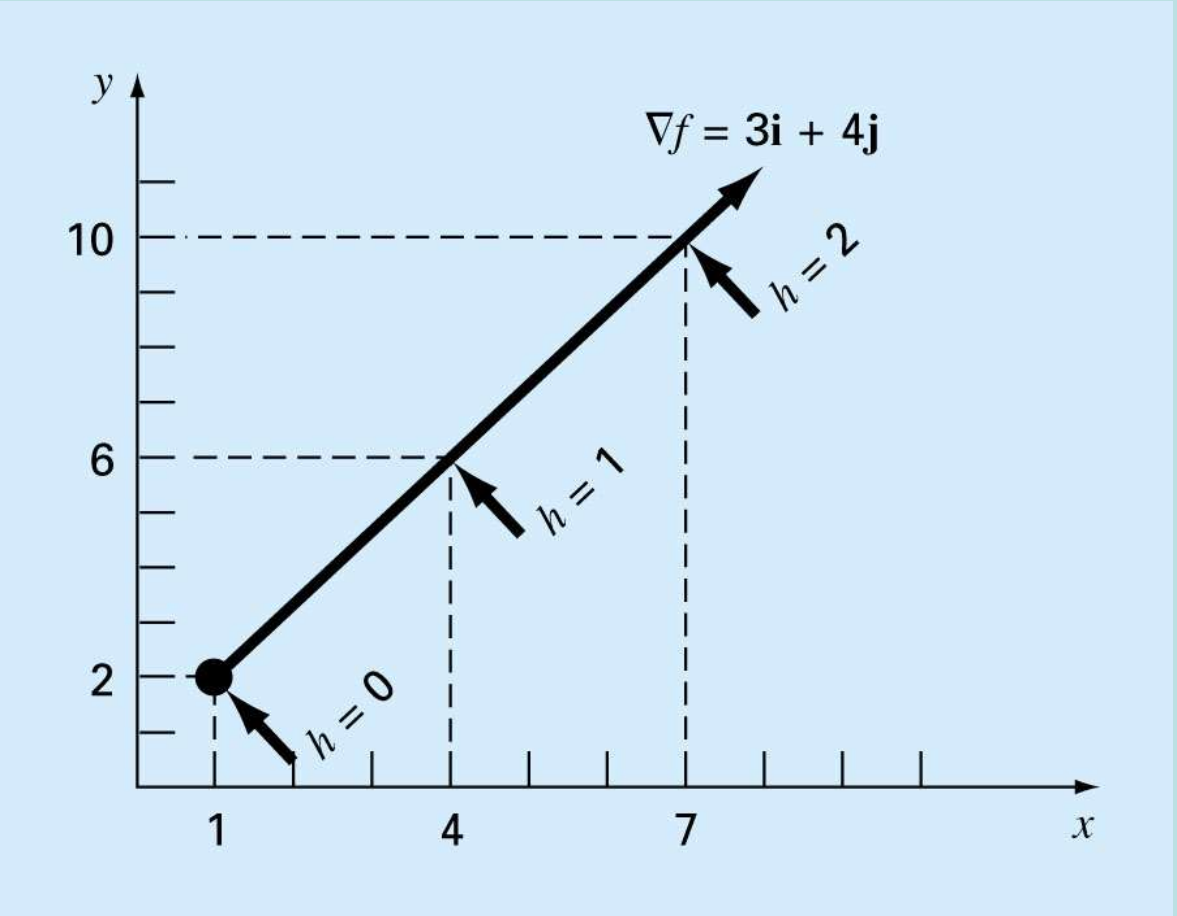
- If $x_o=1$ and $y_o=2$

$$\nabla f = 3\mathrm{i} + 4\mathrm{j}$$

$$x = 1 + 3h$$

$$y = 2 + 4h$$

Figure 14.10



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EXAMPLE 14.3 Developing a 1-D Function Along the Gradient Direction

Problem Statement. Suppose we have the following two-dimensional function:

$$f(x, y) = 2xy + 2x - x^2 - 2y^2$$

Develop a one-dimensional version of this equation along the gradient direction at point $x = -1$ and $y = 1$.

Solution. The partial derivatives can be evaluated at $(-1, 1)$,

$$\frac{\partial f}{\partial x} = 2y + 2 - 2x = 2(1) + 2 - 2(-1) = 6$$

$$\frac{\partial f}{\partial y} = 2x - 4y = 2(-1) - 4(1) = -6$$

Therefore, the gradient vector is

$$\nabla f = 6\mathbf{i} - 6\mathbf{j}$$

To find the maximum, we could search along the gradient direction, that is, along an h axis running along the direction of this vector. The function can be expressed along this axis as

$$f\left(x_0 + \frac{\partial f}{\partial x}h, y_0 + \frac{\partial f}{\partial y}h\right) = f(-1 + 6h, 1 - 6h)$$

$$= 2(-1 + 6h)(1 - 6h) + 2(-1 + 6h) - (-1 + 6h)^2 - 2(1 - 6h)^2$$

where the partial derivatives are evaluated at $x = -1$ and $y = 1$.

By combining terms, we develop a one-dimensional function $g(h)$ that maps $f(x, y)$ along the h axis,

$$g(h) = -180h^2 + 72h - 7$$

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Optimal Steepest Ascent

Problem Statement. Maximize the following function:

$$f(x, y) = 2xy + 2x - x^2 - 2y^2$$

using initial guesses, $x = -1$ and $y = 1$.

Now let us implement steepest ascent. Recall that, at the end of Example 14.3, we had already implemented the initial steps of the problem by generating

$$g(h) = -180h^2 + 72h - 7$$

Now, because this is a simple parabola, we can directly locate the maximum (that is, $h = h^*$) by solving the problem,

$$g'(h^*) = 0$$
$$-360h^* + 72 = 0$$
$$h^* = 0.2$$

This means that if we travel along the h axis, $g(h)$ reaches a minimum value when $h = h^* = 0.2$. This result can be placed back into Eqs. (14.10) and (14.11) to solve for the

(x, y) coordinates corresponding to this point,

$$x = -1 + 6(0.2) = 0.2$$
$$y = 1 - 6(0.2) = -0.2$$

$$x = x_0 + \frac{\partial f}{\partial x}h \qquad (14.10)$$

$$y = y_0 + \frac{\partial f}{\partial y}h \qquad (14.11)$$

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